

LAURA BURNETT, PH.D.

Research Data Scientist | End-to-End Data Pipelines | Multi-Modal Timeseries & Network Analysis | Dashboards & Visualisations

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PROFILE

Research data scientist with ten years' experience turning large, high-dimensional biological data – imaging and video, electrophysiological time series and connectomes (graph databases) – into clear, usable results. Trained neuroscientist who designs and delivers multi-year research programmes and builds production-grade data tools in fast-paced, multidisciplinary teams. Committed to reproducibility and FAIR principles at every stage. Now looking to bring this toolkit to industry, on new, complex biological datasets.

CORE SKILLS

Python (Pandas, NumPy, SciPy, Plotly, Dash) • SQL • Cypher (Neo4j) • HTML, CSS & JavaScript • MATLAB • Git • Experimental Design & Statistical Analysis

DATA SCIENCE & RESEARCH EXPERIENCE

Postdoctoral Research Associate – Data & Analysis

HHMI Janelia Research Campus (Supervisor: [Michael Reiser](#)) 📅 Aug 2023 – May 2026 📍 Ashburn, VA, USA

- **Built and shipped an internal data product** (*neuView*): a Python CLI that queries a Neo4j graph database of >150,000 cells and generates an interactive, open web resource, now deployed and used across the field alongside a landmark *Nature* (2025) study.
- **Designed end-to-end pipelines** for high-throughput screens – synchronised acquisition, cleaning and QC of large video datasets, automated documentation, statistical analysis and reproducible figure generation.
- **Queried and modelled large-scale relational and graph data** in SQL and Cypher; built reusable analysis tooling that was adopted into another group's open-source toolkit.
- **Worked as an engineer in a multidisciplinary team**: shared codebase, Git and pull-request workflow, code review, and documentation written for downstream users rather than authors.

PhD Researcher – Systems Neuroscience

Institute of Science and Technology Austria (Supervisor: [Maximilian Joesch](#)) 📅 Sep 2016 – May 2023 📍 Klosterneuburg, Austria

- **Built quantitative pipelines for large, noisy multimodal data** – video, behavioural and electrophysiological time series analysis and visualisation.
- **Led a multi-year study** on rapid decision-making in autism models, including experimental design, analysis plan, and forming a new collaboration between two research groups, building the experimental rigs and data pipelines from scratch and performing stereotactic surgeries.
- **Synthesised evidence across distinct fields** (*visual neuroscience, decision-making, autism*) to identify overlap and define tractable questions.

Earlier research roles

UCL Institute of Child Health, University of Melbourne & The Florey Institute 📅 2014 – 2016 📍 London, UK & Melbourne, Australia

Four research placements across three institutions, including quantitative analysis of confocal microscopy image data in models of enteric neuropathy and cortical development; led to publications in *Neuron*, *Nature Communications* & *Developmental Biology*.

COMMUNICATION & LEADERSHIP

- **Mentoring & supervision**: mentored 11+ students (graduate, undergraduate and high-school); active member of a women & non-binary mentoring group at HHMI Janelia (since 2023).
- **Speaking & stakeholders**: invited talks at international conferences and symposia (FENS Forum; Champalimaud Centre; BOKU Vienna; Janelia); elected Graduate Student Association Representative, ISTA (2018–2020).
- **Teaching Assistant**: Neuroscience track course at ISTA (2018–2020); included teaching MATLAB to incoming graduate students.
- **Making technical work usable**: led a coding support group and built documentation used across a lab.

OUTREACH

- **STEM outreach volunteer**: active member of HHMI Janelia's RESET group (Raising Excitement for Science, Engineering and Technology) and its "Hour of Code" initiative, teaching elementary-school students to code in Python.
- **Science communicator**: co-founded "Wissen Schafft's" (2021), a multilingual YouTube series explaining viruses and the immune system to the public and tackling misinformation.
- **Workshop leader**: designed and led a day-long session on optical illusions for "Fakebusters", a high-school summer camp (2021 & 2022).
- **Volunteer maths tutor**: Action Tutoring, Year 6 maths (since Aug 2026).

EDUCATION

MSci Natural Sciences - Neuroscience (First Class Honours)

University College London (UCL) 📅 Sep 2015 – May 2016 📍 London, UK

BSc Natural Sciences - Organic chemistry and Biomedical sciences (First Class Honours)

University College London (UCL) 📅 Sep 2012 – May 2014 📍 London, UK

University of Melbourne (study abroad) 📅 July 2014 – July 2015 📍 Melbourne, Australia

PUBLICATIONS

h-index 5 • 397 citations (Google Scholar).

1. Berg S, Beckett IR, Costa M, ... **Burnett LE**, et al. Sexual dimorphism in the complete Drosophila male central nervous system connectome. *Cell*. 2026;189(18):5504–5526.
2. Nern A, Loesche F, Takemura S, **Burnett LE**, et al. Connectome-driven neural inventory of a complete visual system. *Nature*. 2025;641(8065):1225–1237.
3. **Burnett LE**, Koppensteiner P, Symonova O, et al. Shared behavioural impairments in visual perception and place avoidance across different autism models. *PLoS Biology*. 2024;22(6):e3002668.
4. Hao MM, Bergner AJ, Nguyen HTH, Dissanayake P, **Burnett LE**, et al. Role of JNK, MEK and adenylyl cyclase signalling in speed and directionality of enteric neural crest-derived cells. *Developmental Biology*. 2019;455(2):362–368.
5. Beattie R, Postiglione MP, **Burnett LE**, Laukoter S, Streicher C, Pauler FM, et al. Mosaic analysis with double markers reveals distinct sequential functions of Lgl1 in neural stem cells. *Neuron*. 2017;94(3):517–533.
6. McCann CJ, Cooper JE, Natarajan D, Jevans B, **Burnett LE**, Burns AJ, et al. Transplantation of enteric nervous system stem cells rescues nitric oxide synthase deficient mouse colon. *Nature Communications*. 2017;8(1):15937.

FUNDING & AWARDS

- **Travel grant**, Austrian Neuroscience Association, FENS (2022).
- **OeAD research scholarship**: competitive, self-authored proposal (2016).
- **Dean's List**, University of Melbourne (2014).

REFERENCES

Reference contact information is available upon request.